

Intrathecal Pain Program

Frida · single-reservoir morphine + bupivacaine + clonidine admixture, flow-rate flexed to her daily symptom rhythm. The three agents co-treat pain and share one flow rate by design. Spasticity is managed on a separate baclofen therapy (see note).

Medication Mix – 40 mL reservoir

Medication	Reservoir conc.	Load (40 mL)	Role	Avg daily dose
Morphine	0.6 mg/mL	24 mg	Nociceptive / neuropathic pain	~0.29 mg/day
Bupivacaine	6 mg/mL	240 mg	Sensory / pain adjunct	~2.9 mg/day
Clonidine	60 µg/mL	2.4 mg	Neuropathic adjunct	~29 µg/day

Reservoir 40 mL · average flow ~20 µL/h · ~0.49 mL/day consumed · ~2 mL residual.

A full fill lasts ~78 days (~11 weeks). Filled 2 Jun 2026 — next refill due 19 Aug 2026.

Weekday Dosing

Works from home — higher morning dose, sustained daytime coverage, evening boost.

Interval	Flow µL/h	Morphine µg/h	Bupivacaine mg/h	Clonidine µg/h
Night 23:00–06:00	14	8.4	0.084	0.84
Morning 06:00–09:00	30	18	0.180	1.8
Daytime (base) 09:00–18:00	20	12	0.120	1.2
Evening 18:00–23:00	24	14.4	0.144	1.44

Weekend Dosing

Sleeps in — morning peak shifted later and runs longer; night extends to 08:00.

Interval	Flow µL/h	Morphine µg/h	Bupivacaine mg/h	Clonidine µg/h
Night 23:00–08:00	14	8.4	0.084	0.84
Morning 08:00–12:00	30	18	0.180	1.8
Daytime (base) 12:00–18:00	20	12	0.120	1.2
Evening 18:00–23:00	24	14.4	0.144	1.44

Daily totals (both day types): morphine ~0.29 mg · bupivacaine ~2.9 mg · clonidine ~29 µg. All three sit within PACC recommended starting ranges and at a small fraction of the maximum concentrations (3% / 20% / 6% of max) and maximum daily doses.

Spasticity = separate therapy. Baclofen is not in this reservoir. It is delivered by its own intrathecal pump (or orally) and titrated independently, typically ~300–800 µg/day. Co-mixing baclofen with analgesics is uncommon, has limited stability data, and removes the ability to time spasticity control separately from pain.

Illustrative figures for design / prototyping only — not validated prescribing values. Actual pump programming follows the device labelling and the prescriber’s judgement.